Conditional consequences of a Boolean tensor-contraction theorem: threshold circuits and sparse connectivity

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Abstract

Assuming the Boolean tensor-contraction theorem in OpenAI's manuscript *The Exponential Time Hypothesis is False* (25 October 2026), we collect consequences for threshold circuits and sparse graph problems. For an n-input circuit with w wires and a polynomial length integer weight description, put $S = n+w+2$. Its acceptance probability has a deterministic additive approximation, down to error $2^{-S^{9/10}/\log_2(S+16)}$, in $\mathrm{poly}(S)2^{O(S/\sqrt{\log S})}$ time and polynomial space, with unrestricted depth and fan-in. A per-gate theorem combines exact threshold, equality, and modular gadgets with hashed gadgets; it gives exact counting when no gate is hashed. Exact-only assignments allow arbitrary description length, charged polynomially. Applying established algorithms-to-lower-bounds constructions yields an E^{NP} language requiring more than $cN\sqrt{\log N}$ real-weight threshold wires at every sufficiently large length, and a corresponding statement for every fixed constant approximation advantage. For graphs with $m = O(n)$, a sparse representation of the permutation sign and the matrix-tree identity give deterministic $2^{O(n/\sqrt{\log n})}$ algorithms for connectivity optimization, witness recovery, and exact Hamiltonian-cycle counting. Directed and sparse shortest-path-metric variants are included. Every algorithmic and complexity-theoretic conclusion is conditional on the stated input theorem. The accompanying finite computations check the reductions.

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1 Introduction

Theorem D.1 of OpenAI's 25 October 2026 manuscript [14] asserts a deterministic, polynomial-space contraction algorithm for Boolean tensor networks with linearly many edges and maximum degree $\lceil s^{9/10} \rceil$. We denote this assertion by H_{TC}, and state its full interface in Section 2. This paper studies its consequences. The degree bound, the allowed coefficient ring, the cost of evaluating implicit tensors, and exact integer readback are part of every application.

Throughout, $\log_2$ is the binary logarithm, and $g(s) = s/\sqrt{\log_2(s+16)}$. Running times count bit operations, including polynomial factors in the description length. Polynomial space means polynomially many bits. A circuit wire is an occurrence in a gate's input list; arbitrary fan-out is permitted. Circuit depth and fan-in are unrestricted unless a statement explicitly bounds them.

1.1 Statements of the consequences

Theorem 1.1 (Threshold acceptance probability, conditional on H_{TC})**.** *Assume H_{TC}. Fix $a \geq 1$. Let C be a single-output Boolean threshold circuit with n declared inputs, w wires, and integer weights and thresholds in a complete description of length $L \leq S^a$, where $S = n+w+2$. For every*

$$2^{-S^{9/10}/\log_2(S+16)} \leq \varepsilon < 1,$$

there is a deterministic algorithm producing a rational $\widehat{p}$ with

$$\left|\widehat{p} - 2^{-n}|\{x : C(x) = 1\}|\right| \leq \varepsilon$$

in $\mathrm{poly}_a(S+L)2^{O_a(g(S))}$ time and $\mathrm{poly}_a(S+L)$ space. In particular this holds for $\varepsilon \geq S^{-a}$. The algorithm can use the fixed lower endpoint as its target error, so no arbitrary-precision real encoding of ε is required.

The proof is in Sections 3–4. Theorem 4.1 gives the precise per-gate version. It allows THR, exact-THR, and weighted MOD_m gates, as well as small gates with polynomial-time evaluators. Each large gate is assigned either an exact sliced gadget, an exact supplied weight-factorization gadget, or a hashed gadget. Under the explicit wire budget (4.1), the same time and space bounds hold, with error at most the sum of the hashed gates' error budgets. When that set is empty, the answer is the exact number of accepting assignments. In that exact-only case the description length L is arbitrary, with polynomial cost in $S + L$. In particular, exact counting is available when the wire-weighted sum of large gates' parameter bit lengths is $O(S^{19/10})$.

Theorem 1.2 (Real-weight wire lower bounds, conditional on H_{TC})**.** *Assume H_{TC}. There are a constant $c > 0$ and a language $L_{\mathrm{ae}} \in \mathrm{E}^{\mathrm{NP}}$ such that, for every sufficiently large N, every real-weight threshold circuit computing $(L_{\mathrm{ae}})_N$ has more than $cN\sqrt{\log_2 N}$ wires. Consequently there is also such a language with this lower bound at infinitely many lengths. For every fixed $\gamma \in (0, 1/2)$ there are $c_\gamma > 0$ and $L_\gamma \in \mathrm{E}^{\mathrm{NP}}$ such that, for every sufficiently large N, every real-weight threshold circuit with at most $c_\gamma N\sqrt{\log_2 N}$ wires agrees with $(L_\gamma)_N$ on at most a $1/2+\gamma$ fraction of inputs. The circuits have arbitrary depth and fan-in. The language in the last assertion may depend on γ.*

Section 5 proves these statements by substituting the acceptance-probability algorithm into the proofs of known transfer constructions. We also give the separate infinitely-often construction, since it makes the two-stage verification cost explicit.

Theorem 1.3 (Undirected sparse connectivity, conditional on H_{TC})**.** *Assume H_{TC}. Fix $c > 0$. On simple undirected graphs with n vertices and $m \leq cn$ edges, Hamiltonian cycle, Hamiltonian path (with specified or unspecified endpoints), longest simple path, Hamiltonian TSP, Steiner tree, feedback vertex set, connected vertex cover, and connected dominating set admit deterministic decision, optimization, and witness recovery in*

$$\mathrm{poly}_c(n+L)2^{O_c(g(n))} \quad \textit{time and} \quad \mathrm{poly}_c(n+L) \quad \textit{space.} \tag{1.1}$$

The relevant vertex and edge costs may be nonnegative integers of at most $\lceil n^{9/10} \rceil$ bits. One can also exactly count undirected Hamiltonian cycles as edge sets, Hamiltonian paths with specified endpoints, simple paths with specified endpoints and length, and Steiner tree subgraphs (U, F). For connected covers, connected dominating sets, and feedback vertex sets, the proof uses positive weighted counts to decide existence and optimize.

Theorem 1.4 (Directed sparse connectivity, conditional on H_{TC})**.** *Assume H_{TC}. Fix $c > 0$. On digraphs without loops or parallel arcs and with $m \le cn$, the bounds* (1.1) *hold for exact directed Hamiltonian-cycle counting, exact counting of directed paths with specified endpoints and length, longest directed simple path, nonnegative-cost directed Steiner tree, and minimum strongly connected spanning subgraph. They also hold for a minimum strongly connected subgraph containing specified terminals. Directed Steiner arborescences with a fixed root, direction, and terminal set can be counted exactly. Costs have at most $\lceil n^{9/10} \rceil$ bits. Optimization includes recovery of an optimal subgraph.*

Theorem 1.5 (Routing on a sparse base graph, conditional on H_{TC})**.** *Assume H_{TC}. Fix $c > 0$. Given a simple undirected base graph with $m \le cn$, nonnegative integer edge costs of at most $\lceil n^{9/10} \rceil$ bits, a terminal set T, and a required edge set R, a minimum-cost closed walk visiting T and traversing R can be found, or infeasibility detected, within* (1.1)*. This includes graphical TSP, Steiner TSP, and undirected Rural Postman. On a connected base graph, $T = V$ and $R = \varnothing$ give TSP on its shortest-path metric, with an optimal metric tour recovered for $n \ge 3$.*

1.2 Comparison and scope

Table 1 compares exponent scales. In its previous-best column, $2^{\Theta(n)}$ denotes the single-exponential scale of the available generic bounds; different problems and graph subclasses have different constant bases. The cited algorithms specify representative bounds within that scale. The comparison concerns unrestricted depth in the circuit rows and a fixed edge-to-vertex ratio in the graph rows.

Problem	Previous best (exponent scale)	Under H_{TC}
Arbitrary-depth THR CAPP, $w = O(n)$	$2^{\Theta(n)}$ generic enumeration	$2^{O(g(n))}$, error down to $2^{-\Theta(n^{.9}/\log n)}$
Mixed-gate exact counting under (4.1)	$2^n \operatorname{poly}(L)$ enumeration	$\operatorname{poly}(S+L)2^{O(g(S))}$ for arbitrary L if no gate is hashed
Unrestricted-depth real THR wires for explicit E^{NP} languages	Linear scale; depth-restricted bounds differ	$> cN\sqrt{\log N}$ a.e.; fixed-advantage average case
Sparse Hamiltonicity	Single exponential; e.g. randomized $O^*(1.657^n)$ [2]	Deterministic $2^{O(g(n))}$; exact #HC
Sparse paths, Steiner, FVS, connected cover/domination	$2^{\Theta(n)}$ scale; connectivity frameworks [7, 3]	Deterministic $2^{O(g(n))}$, optimization and witnesses
Sparse directed connectivity and sparse-base routing	$2^{\Theta(n)}$ scale (enumerate selected arcs or edge multiplicities)	Deterministic $2^{O(g(n))}$

Table 1: Polynomial factors in the input length are suppressed. Algorithmic comparisons are deterministic unless randomization is marked. The conditional bounds retain polynomial space. The exact-counting rows use the object conventions in the theorem statements.

The polynomial description condition in Theorem 1.1 is a bit length condition. For lower bounds, each real-weight threshold gate has an equivalent polynomial-bit integer representation, with no change in wires. This is an existence assertion, sufficient for guessing circuits; it does not prescribe how to read arbitrary real-number oracles. For an algorithm on a larger wire budget, the displayed bound is in S, and may exceed exhaustive enumeration over the n inputs.

1.3 Related work

OpenAI's manuscript itself derives a circuit lower bound from the fixed-density case of its Theorem 1.1 [14, Corollary 1.4]: there are $L \in \mathrm{E}^{\mathrm{NP}}$ and $c > 0$ such that circuits of fan-in-two Boolean gates computing L need more than $cn\sqrt{\log n}$ noninput gates for infinitely many n. Its proof combines Williams' witness-circuit argument with a local reduction to 3-CNF. Theorem 1.2 uses the stronger input H_{TC} and a different measure. Every fan-in-two Boolean gate has a threshold implementation with at most five wires, so Theorem 1.2 implies a gate lower bound of the same order for fan-in-two circuits, at every sufficiently large length and in the constant-advantage form. In the other direction, the generic simulation of Section 5 replaces a fan-in-r threshold gate by $O(r^2 \log(r+2))$ fan-in-two gates, so Corollary 1.4 does not yield the wire bound of Theorem 1.2 by gate simulation.

Williams' algorithmic method [18] connects savings in circuit analysis to circuit lower bounds. Chen and Williams [6] use PCPs of proximity to reduce the composition overhead in such connections. Bathie and Williams [1, Section 3.2–3.3] give a two-stage PCP/ECC/PCPP construction whose circuit queries are small compositions of proof-bit circuits. The infinitely-often argument here is a routine instantiation of that proof framework in the wire measure. Chen, Lyu, and Williams [4] give the refuter for a hierarchy with sublinear nondeterminism and an XOR amplification based on linear sums. Their proof mechanisms supply the almost-everywhere and constant-advantage arguments below; the polynomial parameters and the per-term wire counts are recalculated explicitly.

Depth-restricted threshold bounds have different quantitative behavior. Impagliazzo, Paturi, and Saks [8] prove that parity requires $n^{1+c\theta^{-d}}$ wires at depth d, for absolute constants $c > 0$ and $\theta \leq 3$. Kane and Williams [10] prove, for a fixed advantage, depth-two lower bounds of order $n^{5/2}/\log^{7/2} n$ wires and $n^{3/2}/\log^3 n$ gates for an explicit function; their depth-three result has a majority output gate. Chen, Tal, and Wang's 2026 manuscript [5, Theorems 1.1–1.2] states $n^{2.5-\varepsilon}$ gate lower bounds in E^{NP} for depth-two linear threshold circuits, via a CAPP algorithm for an XOR of two such circuits. These gate and wire measures and their depth restrictions are retained in the comparison. Li and Yang [11] prove a $3.1n - o(n)$ gate lower bound over the full binary Boolean basis. A binary Boolean gate has a constant-size threshold implementation; the reverse simulation of an arbitrary-weight threshold gate has a larger bit cost, as quantified in Section 5.

The comparison protocol used here follows Nisan's integer-comparison construction [12]. Viola [17] improves the communication for addition and comparison. We use an explicit half-splitting and prime-fingerprinting implementation to keep seed length, local evaluation, and tensor ports in the same ledger. The conversion from a deterministic transcript to local consistency tensors and the union bound over the earliest erroneous gate are standard constructions.

For graph connectivity, Cut&Count [7] combines consistent cuts with isolation to filter disconnected candidates modulo two. Bodlaender, Cygan, Kratsch, and Nederlof [3, Section 4] use determinants and the matrix-tree theorem for deterministic connectivity counting parameterized by treewidth. Björklund's Hamiltonicity algorithm [2] uses a different determinant summation and cancellation. Our graph proofs instantiate the matrix-tree approach using an explicit representation of the sparse permutation sign by linearly many Boolean wires. The resulting scale is $n+m$; no treewidth parameter is substituted for the number of vertices. The $0/1/2$ multiplicity reduction for graphical TSP also appears in the Cut&Count treatment [7, Section 4.10].

2 The contraction interface and encoding conventions

Hypothesis 2.1 (H_{TC}: OpenAI, Theorem D.1)**.** Fix positive constants C_0, A_0, b_0. For an integer $s \geq 2$, let a bipartite multigraph carry one Boolean variable on each edge. There are at most $C_0 s$ edges and a number of vertices bounded by a fixed polynomial in s, on which the constants below may also depend, and its maximum degree is at most $\lceil s^{9/10} \rceil$. If the total graph and tensor description length is $\mathcal{L}$, the tensor entries lie in

$$\mathcal{B}_{P_0,\ell} = (\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad 1 \leq P_0, \ell \leq C_0 s.$$

Uniformly for every vertex and every Boolean port assignment, an entry can be computed from its description in time $(s+\mathcal{L})^{b_0} 2^{A_0 s^{9/10}}$ and space $(s+\mathcal{L})^{b_0}$. Then the contraction

$$Z = \sum_{x \in \{0,1\}^E} \prod_v T_v(x|_{\delta(v)}) \tag{2.1}$$

is computable deterministically in $(s+\mathcal{L})^{O(1)} 2^{O(g(s))}$ bit operations and $(s+\mathcal{L})^{O(1)}$ space. Constants depend only on the fixed bounds just specified. Isolated vertices contribute their scalar entries; $\ell = 1$ is allowed.

This is Theorem D.1 (Boolean tensor contraction) of [14]. All our applications use $\ell = 1$. The only coefficients required are integers modulo a power of two, including negative residues for determinants.

For comparison, write H_{BB} for Theorem 1.1 of [14]: for each fixed $k \geq 2$, exact counting for k-CNF on $n \geq 20$ declared variables and m clauses takes polynomial space and time

$$(n+m)^{O_k(1)} 2^{O_k(n \log\log n / \sqrt{\log n})}.$$

For each fixed D, if $m \leq Dn$, the exponent improves to $O_{k,D}(g(n))$. The quantifier fixes k and D before the input size grows.

The accompanying Lean scope document [13], `lean/docs/378.md`, describes exact counting for fixed k on canonical encodings, in time

$$(L+1)^{O_k(1)} 2^{O_k(n(\log\log n)^2/\sqrt{\log n})}$$

and with polynomially many visited tape cells. Its stated scope is this fixed-k counting assertion. It does not include H_{TC}, nor the sharper arbitrary-density or fixed-density bounds in the October 25 Theorem 1.1. Thus that formalization does not certify the hypothesis used by the results of this paper.

2.1 Boolean wires and unique extensions

A functional gadget has input and output ports and internal Boolean wires. For each assignment to its input ports, exactly one assignment to its internal wires and output ports has nonzero product of local 0/1 entries. A circuit of such gadgets has a unique extension of each original input, by induction in a topological order. Pinning the final output to one then makes its contraction the accepting-input count.

Fan-out is handled by equality trees. For a signal with $d \geq 2$ occurrences, join its ports by a tree of binary and ternary equality vertices; fixing the common signal determines every internal bit. For $d = 1$, attach the free unary tensor $(1, 1)$. A declared input with no occurrence is represented by a wire between two copies of $(1, 1)$, contributing exactly two. In particular, changing a gate to a constant never deletes the factor contributed by an input that thereby becomes unused. These rules also apply when the circuit output is an input or a constant.

A constant gate is evaluated directly and its outgoing uses are fixed by unary pins. A nonconstant gate has at least one input wire, so the number of remaining gates is at most w. Constants, dead gates, and zero-arity scalar factors can be removed from the network description after accounting for their effect and retaining all declared inputs.

Given any finite factor graph, insert a binary equality tensor on each edge. The old and new vertices form a bipartition, edge count doubles, maximum degree does not increase, and the assignment on each old edge has exactly one extension to its two replacements. If the sum of port occurrences is $O(S)$, equality trees and subdivision therefore use $O(S)$ edges and vertices. Distinct occurrences at the same factor are distinct ports. If a two-occurrence signal joins two ports of that factor, subdivision creates parallel edges to the equality vertex. These are allowed by the bipartite multigraph interface of H_{TC}.

2.2 Scaling and integer readback

Suppose an encoding has $C_1 S$ edges, $O(S)$ vertices, degree at most $D_1 S^{9/10}$, precision $P_0 \le C_2 S$, and total description length $\mathcal{L}$. Suppose its local entry programs use fixed polynomial time and space in $S + \mathcal{L}$. Set

$$s = \lceil KS \rceil, \qquad K \ge \max\{1, (2D_1)^{10/9}\}, \tag{2.2}$$

and fix a sufficiently large interface constant C_0. All the bounds in H_{TC} now hold, including its precise maximum-degree condition. Since K is fixed, $g(s) = O(g(S))$. The same reasoning applies to $O(S^{2/3})$ degrees. The resulting time is $(S + \mathcal{L})^{O(1)} 2^{O(g(S))}$ and the space is polynomial in $S + \mathcal{L}$, allowing arbitrary description length. Bounded small instances may be enumerated with polynomial cost in their descriptions.

If a contraction is known, as an integer, to satisfy $0 \le Z < 2^{P_0}$, its least nonnegative residue modulo 2^{P_0} recovers Z exactly. For a circuit with unique extensions, $P_0 = n + 1$ is sufficient, since $0 \le Z \le 2^n$. For a signed graph encoding the bound is proved after cancellation, before choosing P_0. Division by a known multiplicity is performed on this recovered integer.

Part I
Threshold circuits

3 Comparison protocols as bounded-degree relations

We give the comparison implementation used in the threshold gadgets. It is the half-splitting comparison underlying Nisan's protocol [12]; explicit parameters are useful for applying H_{TC}. Fix the circuit scale S, and suppose $t \le S$ parties hold nonnegative integers X_i, each of bit length polynomial in S. The public target T has the same bit bound. We compare $\sum_i X_i > T$. Targets outside the possible sum range are handled by public constant answers. Pad the bit width to a power of two B' and put $J = \log_2 B' = O_a(\log S)$ when the input description is at most S^a bits.

3.1 The half-splitting step

For a current even width B, put $d = B/2$ and write

$$X_i = 2^d H_i + L_i, \quad T = 2^d T_H + T_L, \quad 0 \le L_i, T_L < 2^d.$$

If $\sum_i H_i = T_H - h$ for some $h \in \{0, \dots, t-1\}$, then the comparison is equivalent to

$$\sum_i L_i > T_L + h 2^d. \tag{3.1}$$

If there is no such h, put $u = \sum_i H_i - T_H$. Then either $u \geq 1$ or $u \leq -t$. In the former case the original comparison is true, since $2^d u + \sum_i L_i - T_L \geq 1$; in the latter case it is false. Both answers equal the answer to $\sum_i H_i > T_H$. Thus one may recurse on that high-half comparison. The range for h includes $t - 1$.

Let $\delta = 2^{-h_0}$, with integer $h_0 \geq 1$. Choose the least power of two K such that

$$K \geq 4JS(B' + \lceil \log_2 S \rceil + 3)\, 2^{h_0}. \tag{3.2}$$

Use the K smallest primes greater than S. At each level a fresh $\log_2 K$-bit seed chooses one such prime p. The parties broadcast $H_i \bmod p$, and public arithmetic computes the least nonnegative residue

$$h' = T_H - \sum_i (H_i \bmod p) \pmod p.$$

If $h' < t$, follow (3.1) with $h = h'$; otherwise follow the high-half comparison. Since $p > t$, a genuine h is always returned correctly. When no genuine h exists, an incorrect choice requires p to divide one of the nonzero integers

$$\sum_i H_i - T_H + j, \qquad 0 \leq j < t.$$

After the range checks, each has bit length at most $B' + \lceil \log_2 S \rceil + 3$. An integer of that bit length has at most that many distinct prime factors. Consequently the conditional error at a level is at most $\delta/(4J)$, even after fixing all earlier messages and seeds. A union bound on the earliest erroneous level gives error at most δ. At width one the parties send their remaining bits and compare exactly.

The prime bound of Rosser and Schoenfeld [15] implies that the largest prime used has

$$\pi = O_a(h_0 + \log S) \tag{3.3}$$

bits. Each seed is only a prime index. For a particular index, enumerate integers and use trial division until reaching that prime. The polynomial upper bound on the indexed prime implies $2^{O_a(h_0 + \log S)}$ time and polynomial bit space. This construction stores the selected primes, rather than the entire family. The total seed length is

$$R = J \log_2 K = O_a(h_0 \log S + \log^2 S). \tag{3.4}$$

Fixed-width message frames, range checks, and forced zero padding after an early stop make every transcript canonical.

3.2 Two tensor realizations

Fix the seed. If a party reads at most b original input bits and the full transcript has C bits, make one factor for each party, reading its private input and a copy of the transcript. The factor checks only that party's messages, each against the earlier prefix. A referee factor checks public control, formatting, padding, and the output. Sharing transcript bits by equality trees gives

$$M = O(tb + tC + C + 1), \qquad \Delta \leq \max\{b + C, C + 1, 3\}. \tag{3.5}$$

For fixed private inputs, the initial message is determined. Inductively each next message, stop flag, and final output is determined; the checks force exactly this execution. Hence the gadget is a functional relation even for a seed at which the comparison is wrong.

For higher precision we use the addition-specific compact realization. Encode each public branch (a value of h or "no such value") in $\lambda = \lceil \log_2(t+1) \rceil$ bits. The factor for party i reads

its original input, the public branch history, its own J residues, and its final remaining bit. It replays its local state and checks these messages. It reads no other party's residues. At each level a public factor uses earlier history to produce the target residue and the stop flag. A fan-in-two Boolean circuit adds the t residues modulo the known prime, subtracts from the target modulo that prime, and compares the result with t. Addition of two π-bit residues, comparison with p, subtraction, and multiplexing have $O(\pi)$ gates and wires. Therefore this public arithmetic costs $O(t\pi)$ wires per level. A final factor checks the output from the t last bits and the public history.

Counting original input ports, residues, arithmetic wires, and history copies gives

$$M = O\big(tb + tJ\pi + tJ^2\lambda + J\pi + 1\big), \tag{3.6}$$

$$\Delta = O\big(\max\{b + J(\pi+\lambda),\, \pi + J\lambda,\, t + J\lambda,\, 3\}\big). \tag{3.7}$$

The history-copy bound is a convenient overestimate of the actual port count. Given the inputs, earlier branches determine current messages; those messages determine the next branch. This induction also proves uniqueness for the compact realization. All local entry programs use polynomial-time integer arithmetic on the original weights and the selected primes.

3.3 Shared seeds and the inverse-polynomial CAPP bound

For a circuit, suppose gate v is replaced by $f_{v,\rho}$ satisfying

$$\Pr_{\rho}[f_{v,\rho}(y) \neq f_v(y)] \leq \eta_v \quad \text{for each fixed local input } y.$$

Every gate may read a prefix of the same uniform seed. For an original input x, evaluate the original circuit and let $y_v(x)$ be its fixed local input at gate v. If the replaced circuit disagrees, its earliest erroneous gate has input exactly $y_v(x)$. Therefore

$$\Pr_{\rho}[C_\rho(x) \neq C(x)] \leq \sum_v \Pr_{\rho}[f_{v,\rho}(y_v(x)) \neq f_v(y_v(x))] \leq \sum_v \eta_v. \tag{3.8}$$

No independence between gates is needed.

For clarity, the inverse-polynomial case already follows from the full-transcript realization. Put $b = \lceil S^{2/3} \rceil$ and leave smaller gates exact. For a large threshold gate, replace negative weights by literals: with $u_j = |w_j|$ and $\xi_j = x_j$ for $w_j \geq 0$, $\xi_j = 1 - x_j$ otherwise,

$$\sum_j w_j x_j \geq \theta \quad \Longleftrightarrow \quad \sum_j u_j \xi_j > \theta + \sum_{w_j<0} |w_j| - 1.$$

Party i computes the nonnegative sum in one block of at most b inputs. Give each large gate error $\eta = \varepsilon/(2S)$, using the protocol with a dyadic error at most this value. If $\varepsilon \geq S^{-a}$, then $R = O_a(\log^2 S)$ and a gate with fan-in r_v has $t_v \leq 2r_v/b$ parties and transcript $C_v = O_a(t_v \log^2 S)$. The total additional edges in (3.5) satisfy

$$\sum_{r_v > b} O(t_v C_v) \leq O_a\left(\frac{\log^2 S}{b^2} \sum_v r_v^2\right) \leq O_a(S^{2/3} \log^2 S) = o(S).$$

The degree is at most $S^{2/3} + O_a(S^{1/3} \log^2 S) + O(1) = o(S^{9/10})$. For each fixed seed, the circuit has a unique-extension network and hence an exact count Z_ρ from H_{TC} with $P_0 = n + 1$. Enumerating all seeds gives the rational

$$\widehat{p} = \frac{\sum_\rho Z_\rho}{2^{n+R}}. \tag{3.9}$$

There are at most S gates, so (3.8) bounds the error by ε. The seed enumeration is absorbed by $2^{O_a(g(S))}$, and the accumulator has $n + R + 1$ bits. This proves the inverse-polynomial assertion of Theorem 1.1 under H_{TC}. The compact realization supplies its stated endpoint in the next section.

4 Per-gate exact and approximate counting

Let C be a single-output Boolean DAG. Its gates have ordered input lists, possibly with repetitions. Put $S = n + w + 2$ and $b = \lceil S^{9/10} \rceil$. A gate of fan-in $r_v > b$ is large; all other gates are small. The allowed types are

$$\text{THR} : \left[\sum_j w_j y_j \geq \theta \right],$$

$$\text{exact-THR} : \left[\sum_j w_j y_j = \theta \right],$$

$$\text{MOD}_m : \left[\sum_j w_j y_j \equiv \theta \pmod{m} \right],$$

where all parameters are integers and $m \geq 1$. Let L be the length of the complete input description, including the mode assignment and the certificates below, and fix $a \geq 1$. A small gate can instead be an arbitrary Boolean function supplied by an evaluator. When any gate is hashed, require $L \leq S^a$ and evaluator time at most S^a. For an exact-only assignment, L may be arbitrary and evaluator time at most $(S + L)^a$ is sufficient.

For a large threshold or exact-threshold gate, let B_v be the maximum bit length of $|w_j|$, at least one. For a modular gate let $B_v = \max\{1, \text{bits}(m-1)\}$. Each large gate receives one of the following modes. In the factorization mode, preprocessing verifies the dimensions and entry bounds of the certificate and checks each row of $w = C_v g_v$ by integer arithmetic. A malformed certificate is rejected; the verification costs polynomial time in $S + L$.

Mode	Gate types	Cost σ_v	Error
Exact slices	All three	B_v	0
Exact factorization	All three	$\kappa_v = d_v(\beta_v + \lceil \log_2(r_v + 1) \rceil + 2)$	0
Hashed comparison	THR	$h_v \lceil \log_2 S \rceil$, $h_v \geq 1$	2^{-h_v}
Hashed equality	exact-THR, MOD_m	h_v, $h_v \geq 1$	2^{-h_v}

For factorization, C_v is an $r_v \times d_v$ integer matrix with entries of absolute value less than 2^{β_v}, $\beta_v \geq 1$, and $g_v \in \mathbb{Z}^{d_v}$. Its coordinates can be large; their full bit length is charged to L.

Theorem 4.1 (Per-gate counting, conditional on H_{TC})**.** *Assume H_{TC}. Fix $a, c \geq 1$. Suppose the assigned modes satisfy*

$$\sum_{v \text{ large}} r_v \left\lceil \frac{\sigma_v}{b} \right\rceil \leq cS, \qquad \sigma_v \leq cb \quad \text{for every factorized or hashed large gate.} \tag{4.1}$$

Let $\mathcal{H}$ be the set of hashed gates. If $\mathcal{H} \neq \varnothing$, under $L \leq S^a$, in $\text{poly}_{a,c}(S) 2^{O_{a,c}(g(S))}$ time and polynomial space a deterministic algorithm outputs a rational $\widehat{p}$ with

$$\left| \widehat{p} - \Pr_x[C(x) = 1] \right| \leq \sum_{v \in \mathcal{H}} 2^{-h_v}.$$

If $\mathcal{H} = \varnothing$, the restriction $L \leq S^a$ is unnecessary: the algorithm outputs the exact accepting-input count in $\text{poly}_{a,c}(S + L) 2^{O_{a,c}(g(S))}$ time and $\text{poly}_{a,c}(S + L)$ space, under the exact-only evaluator condition above.

4.1 Exact sliced gadgets

A small gate has the single factor $[z = f(y)]$ of degree $r_v + 1$. Now fix a large gate with fan-in r, width B, and let $K = \lceil B/b \rceil$. For THR or exact-THR, replace signed weights by $u_j = |w_j|$ and the corresponding literal ξ_j as above. Write $T = \theta + \sum_{w_j<0} |w_j|$ and $\Sigma = \sum_j u_j \xi_j$. We have $0 \le \Sigma < r2^B$. For a threshold gate clamp $T < 0$ to zero; for an exact-threshold gate replace $T < 0$ by $r2^B$. In either case clamp $T > r2^B$ to $r2^B$. These operations preserve the Boolean function. For a modular gate, replace w_j by $u_j = w_j \bmod m$ and θ by $\theta' = \theta \bmod m$, with $\xi_j = y_j$. Then $u_j < 2^B$ and $0 \le \Sigma < rm$.

Decompose each u_j into base-2^b digits $u_j^{(k)}$, $0 \le k < K$. For each digit position compute

$$Y^{(k)} = \sum_j u_j^{(k)} \xi_j, \qquad \Sigma = \sum_{k=0}^{K-1} 2^{kb} Y^{(k)}.$$

Partition the r inputs into $t = \lceil r/b \rceil \le 2r/b$ blocks. A chain for each k reads one block at a time and checks the exact partial-sum update, passing a state of $\mu = b + \lceil \log_2(r+1) \rceil$ bits. Each input is read K times. The chain has $O(r)$ wires per slice, degree $b + 2\mu$, and unique states on every original input.

The final arithmetic relates the slice sums by short carries. For nonnegative constants A, T' occurring below put

$$K^* = \left\lceil \frac{B + \lceil \log_2(r+2) \rceil}{b} \right\rceil.$$

They are less than $2^{K^* b}$. Let A_k, T'_k be their base-2^b digits and $Y^{(k)} = 0$ for $k \ge K$. Starting with $c_{-1} = 0, e_{-1} = 1$, enforce

$$\begin{aligned} z_k &= Y^{(k)} + A_k - T'_k + c_{k-1}, \\ c_k &= \left\lfloor z_k / 2^b \right\rfloor, \qquad\qquad e_k = e_{k-1} \wedge [z_k \equiv 0 \pmod{2^b}]. \end{aligned} \tag{4.2}$$

Induction shows

$$\Sigma + A - T' = c_{K^*-1} 2^{K^* b} + R_0, \qquad 0 \le R_0 < 2^{K^* b}.$$

Thus the difference is nonnegative exactly when the last carry is nonnegative, and it is zero exactly when that carry is zero and the last zero flag is one. Every carry lies in $[-1, r+1]$: if the previous one does, then $-2^b \le z_k \le (r+1)2^b$. An offset encoding therefore uses $\lceil \log_2(r+3) \rceil$ bits. With $A = 0, T' = T$ the last carry and flag give the required THR or exact-THR output.

For MOD_m, put $A = m - \theta'$ and introduce a quotient code $q \in \{0, \dots, r\}$. Compute two instances of (4.2), with $T' = qm$ and $T' = (q+1)m$. The terminal factor requires

$$0 \le \Sigma + (m - \theta') - qm < m$$

and outputs one exactly when its left side is zero. There is precisely one legal q, namely $\lfloor (\Sigma + m - \theta')/m \rfloor$; invalid binary codes are rejected. Every carry factor can read the short q code and compute its needed digits locally. The same quotient gadget includes $m = 1$ without a separate case.

The slice chains, carries, quotient copies, and output use $O(rK)$ wires, degree $O(b + \log S)$, and polynomial-time local entry programs. The bound $K^* \le K + 1$ holds for sufficiently large S; more generally $K^* = K + O(1)$ suffices. The $O(K \log S)$ carry and quotient wires are absorbed by $O(Kr)$ because $r > b$. All internal states and the output are unique once the input is fixed. This establishes the exact sliced mode.

4.2 Exact factorization gadgets

For $w = Cg$, write $\kappa = \kappa_v$ for this gate's state width. The information to transmit across blocks is a vector, rather than the weighted sum itself. After block i store

$$Y_i = \sum_{j \text{ in blocks } 1,\dots,i} y_j C_{j,\cdot} \in \mathbb{Z}^d.$$

Each coordinate has absolute value less than $r2^\beta$. A fixed signed code of $\beta+\lceil\log_2(r+1)\rceil+2$ bits suffices, so the complete vector uses κ bits. Block factors enforce the updates. The last factor checks the desired comparison, equality, or congruence of $g \cdot Y_t$ with θ. The supplied identity $w = Cg$ proves correctness. Updates and output are unique. Before equality expansion and subdivision, the logical wire count satisfies

$$\begin{aligned} r + t\kappa + 1 &\le r + (r/b + 1)\kappa + 1 \\ &\le r + cr + cb + 1 \le (2c + 2)r, \end{aligned}$$

using $t = \lceil r/b \rceil \le r/b + 1$, $\kappa \le cb$, and $r > b$. Thus there are $O_c(r)$ wires for every fixed $c \ge 1$, and degree at most $b + 2\kappa + O(1) = O_c(b)$ when $\kappa \le cb$.

For example, if a gate's weights have at most d distinct values, one may take C to be their indicator matrix and g the list of values. This gives $\beta = 1$ and $\kappa = O(d \log S)$ independently of the bit lengths of those values. The factorization is part of the input; no search for an optimal one is used.

4.3 Hashed equality and congruence

For an exact-threshold gate set $\Sigma = \sum_j w_j y_j$ and $\mathcal{T} = \{\theta\}$. For a modular gate use reduced weights and the target set

$$\mathcal{T} = \{\theta' + qm : 0 \le q < r\}.$$

Because $\Sigma \in [0, rm)$, membership in this set is exactly the original modular predicate. Given a prime $p > S$, replace either predicate by

$$f_p(y) = [\Sigma \bmod p \in \mathcal{T} \bmod p].$$

Every true input remains true. For a false input, a false positive requires a prime divisor of some nonzero $\Sigma - \tau$, $\tau \in \mathcal{T}$. Each such integer has bit length at most $S^{\lceil a \rceil} + 2\lceil \log_2 S \rceil + 3$. Put

$$N_v = r(S^{\lceil a \rceil} + 2\lceil \log_2 S \rceil + 3)$$

and let K_v be the least power of two at least $N_v 2^{h_v}$. Selecting among the K_v smallest primes greater than S gives error at most 2^{-h_v}. Their common bit bound is $\pi_v = h_v + O_a(\log S)$.

A block chain passes the partial sum modulo p in π_v bits. Its last factor tests membership in the at-most-r reduced targets. Reject residue codes outside $[0, p)$. For each fixed prime this is a unique-extension circuit relation, with $O_{a,c}(r)$ wires and degree $O_{a,c}(b)$ if $h_v \le cb$. In particular the end-state contribution is bounded by $O_{a,c}(r)$; it need not be at most r when $c > 1$.

4.4 Hashed comparisons

For a large THR gate, use the compact comparison realization (3.6)–(3.7), grouping b input occurrences per party and choosing error 2^{-h_v}. Its parameters satisfy

$$J = O_a(\log S), \quad \pi_v = O_a(h_v + \log S), \quad \lambda = O(\log S), \quad t \le 2r/b.$$

If $h_v\lceil \log_2 S\rceil \le cb$, then $J\pi_v = O_{a,c}(b)$ and $J^2\lambda = o(b)$. The edge count is $O_{a,c}(r)$ and the degree is $O_{a,c}(b)$. For the terminal $J\pi_v$ term, use $r > b$. The seed length is $O_a(h_v \log S + \log^2 S) = O_{a,c}(b)$. The comparison's pointwise error and the functional property are the ones proved in Section 3.

4.5 Assembling the network

Proof of Theorem 4.1. For every hashed gate use the required prefix of a common seed of length R, where R is the maximum of the gate seed lengths and is zero if no gate is hashed. Both kinds of hashing above give $R = O_{a,c}(S^{9/10})$. A uniform seed has uniform prefixes, so each gate retains its own error guarantee. Generate the selected primes by index, as described above. Per seed the extra time is $2^{O_{a,c}(S^{9/10})}$ and the space is polynomial.

Replace each gate by its gadget. Connect every occurrence of each original signal by the equality construction of Section 2, including a factor two for each unused input, and pin the final output. Internal signals read by several factors, including the MOD quotient code and the comparison protocol's branch history, also use equality trees; these copies are already included in the gadget port counts. For a fixed seed the resulting network counts precisely the accepting assignments of C_ρ. Its edges, including equality and bipartite subdivision, are at most

$$O_{a,c}\left(n + w + \sum_{v \text{ large}} r_v\lceil \sigma_v/b\rceil\right) = O_{a,c}(S).$$

Its degree is at most $D_{a,c}S^{9/10}$ for a fixed $D_{a,c}$ and its vertex count is $O_{a,c}(S)$. Its description length and entry costs are polynomial in $S + L$, hence polynomial in S in the hashed case. Use (2.2) with $s = \lceil K_{a,c}S\rceil$, $P_0 = n + 1$, and $\ell = 1$. All the hypotheses of H_{TC} hold.

Recover the count $Z_\rho \in [0, 2^n]$ exactly. If $R = 0$, this is the answer. Otherwise form (3.9). Applying (3.8) and averaging over x gives the stated error. The number of seeds and the prime-generation cost together add $2^{O_{a,c}(S^{9/10})}$, absorbed by $2^{O_{a,c}(g(S))}$. All calls are serial, and the accumulator uses $n + R + 1$ bits.

In the exact-only case there are no primes to generate and $R = 0$. All arithmetic on weights, thresholds, moduli, and supplied certificates uses polynomial time and space in $S+L$. The edge, degree, vertex, and precision bounds above depend on the mode budget, not on an upper bound for L. Retain the original input in the tensor description, so its length $\mathcal{L}$ satisfies $L \le \mathcal{L} \le \mathrm{poly}_{a,c}(S+L)$. Thus the local evaluation bound is also polynomial in $S+\mathcal{L}$, and the same H_{TC} call gives the stated arbitrary-description bound. □

Proof of Theorem 1.1, including high precision. Use Theorem 4.1 with the fixed description exponent $\lceil a\rceil + 2$ to include generated mode parameters; all resulting constants still depend only on a. Assign hashed-comparison mode to every large gate and choose

$$h_v = \lceil \log_2(2S/\varepsilon)\rceil .$$

Then $\sum_v 2^{-h_v} \le \varepsilon/2$, and at the lower endpoint $h_v\lceil \log_2 S\rceil \le 2b$ for all sufficiently large S. Each summand in (4.1) is at most $2r_v$; taking a fixed $c \ge 2$ proves the required budget. For a parameter-free endpoint implementation one can use the integer target

$$h_v = \left\lceil \frac{\lceil S^{9/10}\rceil}{\lfloor \log_2(S+16)\rfloor} \right\rceil + \lceil \log_2(2S)\rceil + 2,$$

which gives at least the requested accuracy and still obeys the same constant budget. Finite small S are enumerated. Apply Theorem 4.1. □

Corollary 4.2 (Exact specializations, conditional on H_{TC})**.** *Assume H_{TC} and the exact-only evaluator condition above; let L be the arbitrary complete description length. If all large gates are assigned exact slices and*

$$\sum_{v \text{ large}} r_v B_v \le c_1 S^{19/10}$$

for a fixed c_1, exact counting takes $\mathrm{poly}_{a,c_1}(S+L)2^{O_{a,c_1}(g(S))}$ time and $\mathrm{poly}_{a,c_1}(S+L)$ space. The same holds when every large gate has a supplied factorization with $\kappa_v \le c_1 b$. Small gates may have arbitrary parameters with their full lengths charged to L. In particular, this recovers exact counting for bounded-fan-in circuits at the allowed degree scale and for large gates whose weight bit lengths are $O(S^{9/10})$; the modular sliced mode also allows modulus bit lengths $O(S^{9/10})$.

Proof. For slicing use $\lceil B_v/b \rceil \le 1 + B_v/b$ and $\sum_v r_v \le w < S$. For factorization use $\lceil \kappa_v/b \rceil \le \lceil c_1 \rceil$. Theorem 4.1 has no hashed gate in either case. □

4.6 Relations between the specializations

For integer weights, an equality gate has the identity

$$\left[\sum_j w_j y_j = \theta\right] = \left[\sum_j w_j y_j \ge \theta\right] \wedge \neg \left[\sum_j w_j y_j \ge \theta + 1\right]. \tag{4.3}$$

This uses the same weight vector in two threshold gates and only a constant-factor increase in wires. Ordinary unweighted MOD gates of fixed modulus likewise have $O(r)$-wire Boolean, and hence threshold, simulations: pass the running residue in a constant number of bits. They therefore fit the threshold-circuit lower-bound framework of Section 5. For a fixed modulus, arbitrary weights can also be reduced modulo that modulus before this simulation. The modular cases requiring the bit-length or certificate accounting here are predicates with growing modulus and potentially large reduced weights.

The sliced budget is an aggregate condition even for pure THR circuits, whereas the factorization condition bounds each supplied state width κ_v. For example, a large gate with $r_v = \Theta(S^{.95})$ and $B_v = \Theta(S^{.94})$ contributes $\Theta(S^{1.89})$ to the sliced sufficient condition, while its one-dimensional certificate $C = w$, $g = 1$ has width $\kappa_v = B_v + O(\log S)$, above the factorization cap. A shorter certificate is a separate property of its weights. Thus the two sufficient conditions are not identified, including on pure THR circuits. With $L \le S^a$ they are specializations of the polynomial-description statement; their arbitrary-L versions use its exact-only clause.

5 From acceptance probability to wire lower bounds

Write THR$[s]$ for real-weight, arbitrary-depth threshold circuits with at most s wires. Fixing or negating an input changes weights and thresholds without increasing wires; output negation replaces $[w \cdot y \ge \theta]$ by $[-w \cdot y \ge 1 - \theta]$ once integer weights are chosen. Two-input AND, OR, and XOR have constant-wire implementations, so an r-bit parity has $O(r)$ wires. After a projection fixes every input of a gate, evaluate that gate and replace it by constant unary tensors, preserving all remaining declared inputs and adding no wires.

5.1 Finite descriptions and the transfer ingredients

A threshold function on r bits has integer weights and threshold of $O(r \log(r+2))$ bits. Here is the existence proof. Let $D = r + 1$ and for each x put $a_x = (2f(x) - 1)(x, -1)$. Move the

original threshold slightly downward to make all margins strict, and scale to obtain $a_x z \geq 1$ for all x. The rows have rank D, since the rows at $0, e_1, \ldots, e_r$ do. A feasible point with maximal rank among its tight rows has D independent tight rows: otherwise move in a nonzero direction orthogonal to them, choosing its sign to decrease another inequality, until a new row becomes tight. This increases the rank. For the resulting invertible matrix B, Cramer's rule applied to $Bz = \mathbf{1}$ and Hadamard's inequality bound every numerator and denominator by $D^{D/2}$. Multiplication by $|\det B|$ gives the asserted integer representation. Constant functions are handled separately.

Thus a w-wire threshold circuit has a normalized description of $O((n+w)^2 \log(n+w+2))$ bits. This remains true after projections. A normalized fan-in-r gate can also be simulated with $O(r^2 \log(r+2))$ fan-in-two Boolean gates: form the weighted sum with binary addition and compare. A circuit with $w = O(t\sqrt{\log t})$ and $n = O(t)$ consequently has ordinary Boolean size at most t^3 for all sufficiently large t.

We use the following published ingredients in their specified forms. They are quoted tools inside the proofs, and do not add an unproved algorithmic assumption to H_{TC}. All fixed probability thresholds used by a machine are chosen rational, weakening completeness and soundness bounds within their positive gaps when necessary. Thus the acceptance comparisons have finite encodings.

1. The unary nondeterministic time hierarchy supplies $H \in \mathrm{NTIME}(2^t) \setminus \mathrm{NTIME}(2^{t/2})$ [18, Theorem 2.1 (full version)].

2. A succinct PCP for such computations has perfect completeness, constant soundness $\beta < 1$, proof and randomness indices padded to $q = t + O(\log t)$, and polynomial-size query and verification circuits. For the refuter computation of time $T(n)$ we use the stronger soundness n^{-10} and $q = \log_2 T(n) + O(\log \log T(n))$ from [4, Lemma 3.10]. Its queries are projections. The projection query lists and the verifier's fixed-width CNF give fixed polynomial bounds in q on the composed verifier size; their construction time may additionally be polynomial in the original input length.

3. A constant-rate linear error-correcting code Enc with polynomial-time decoder Dec has a fixed decoding radius. The Circuit-Eval PCPP in [6, Lemma 24], used in [1, Lemma 3.6], has perfect completeness, constant soundness $\sigma < 1$ at a fixed proximity, polynomial proof length and proof construction, and constraints that are signed ORs of three queried bits.

4. The refuter in [4, Theorem 4.6] applies to any time-constructible $n \leq T(n) \leq 2^{\mathrm{poly}(n)}$. It supplies a language $A^T_{\mathrm{FS}} \in \mathrm{NTIME}[T(n)]$ and, for each fixed nondeterministic machine M running in $o(T(n))$ time with at most $n/10$ guessed bits, finds for every sufficiently large n an input $z \in \{0,1\}^n$ with $M(z) \neq A^T_{\mathrm{FS}}(z)$, in $\mathrm{poly}(T(n))$ time with a SAT oracle.

Fix once and for all the algorithm supplied by H_{TC}; it is part of the finite description of each cheating machine below. The languages constructed from their refuters are then fixed languages. For each fixed description exponent a, fix an exponent constant $A_5(a)$ such that the inverse-polynomial CAPP algorithm takes

$$\mathrm{poly}_a(S) 2^{A_5(a) g(S)}$$

time on normalized descriptions of length at most S^a, with error at least S^{-a}. This constant is fixed before choosing wire-bound constants.

All wire caps below can be made computable integer functions. For a fixed positive rational d, use

$$B_d(r) = \left\lfloor 2d \ \mathrm{isqrt}\big(r^2 \max\{1, \lfloor \log_2 r \rfloor\}\big) \right\rfloor = (2d + o(1)) r \sqrt{\log_2 r}. \tag{5.1}$$

The floor is computed using rational integer arithmetic. Every candidate circuit is padded to a predetermined polynomial description length, and invalid topologies or out-of-range codes are rejected.

5.2 The infinitely-often construction

Assume H_{TC}. Apply the succinct PCP to the unary hierarchy language H. Fix a common nondecreasing index bound $q(t) = t + O(\log t)$, and pad proof and randomness indices to this bound. Extra random bits are ignored and unused proof bits are zero. The lexicographically least proof accepted on all random strings, if it exists, defines a proof-bit function $W_1(t, x)$; otherwise set that function to zero. The full verifier has size $2^{O(t)}$ when all random strings are written out. Prefix SAT queries recover its lexicographically least proof in $2^{O(t)}$ time with an NP oracle.

Fix a small positive rational d. Define $W_2(t, x, j)$ as follows. Find the lexicographically least normalized integer threshold circuit F_t of at most $B_d(q)$ wires whose truth table is a fully accepted PCP proof. This search is in $2^{O(t)}$ time with SAT queries: for each candidate description, write out the verification at every one of the 2^q random strings, and regard their conjunction as an NP condition on the description bits. Let $G_t(x)$ be rejection by the original PCP using this circuit, and put

$$D_t(y) = 1 - G_t(\mathrm{Dec}(y)).$$

The verifier G_t is an ordinary Boolean circuit. If an exact-threshold or MOD gate is used in such a verifier with polynomial-bit parameters, expand it into binary arithmetic and comparison circuits of polynomial size before applying the PCPP. For every x with $D_t(\mathrm{Enc}(x)) = 1$, take the canonical PCPP proof from its polynomial-time constructor. Its jth bit is $W_2(t, x, j)$. If no F_t exists, or for an invalid index, output zero. Proof lengths are padded to one fixed t^u bound covering all legal candidates, so j has $\ell(t) = O(\log t)$ bits.

Both functions belong to E^{NP} when inputs include a self-delimiting t header. Before doing any exponential work, check that t is at most the full input length and that the payload length is correct. Form a single language $L_\star$ by a bit selecting W_1 or W_2. Its legal lengths are $t + O(\log t)$, with possibly different padding for the two branches. If $L_\star$ had circuits of at most $dN\sqrt{\log_2 N}$ wires at every sufficiently large N, projection would give circuits for both bit functions within $B_d(q)$.

A fast nondeterministic simulator for $H(1^t)$ guesses F and a universal proof-bit circuit $T(x, j)$, each of at most $B_d(q)$ wires. Generate the PCPP constraints $H_1, \ldots, H_m$ of the associated D. Replace every code query by its parity on x and every proof query by the corresponding fixed-j projection of the *same* T. The resulting threshold circuit $J_i(x)$ satisfies

$$w(J_i) \le 3B_d(q) + O(q) \le 6d\, q\sqrt{\log_2 q} + O(q). \tag{5.2}$$

For example, using six wires per two-input XOR gives the explicit bound $3B_d(q) + 18q + 3$. The polynomial-size original verifier is used to generate the PCPP; it is not copied into J_i.

If $H(1^t) = 1$, the assumed small circuit for W_1 makes a legal F_t exist; the small circuit for W_2 supplies T, and every J_i is identically one. If $H(1^t) = 0$, every F makes $G(x) = 1$ on at least $1 - \beta$ of the inputs. For such x, every word within the code's decoding radius of $\mathrm{Enc}(x)$ decodes to x, so is rejected by D. The PCPP soundness then makes at least $1 - \sigma$ of the constraints fail for *any* proof at that x. Averaging gives some i with

$$\Pr_x[J_i(x) = 0] \ge \tau = (1 - \beta)(1 - \sigma) > 0.$$

CAPP error $1/S_i < \tau/4$ distinguishes these cases by testing all estimates against $1 - \tau/2$.

Integer normalization ensures $L_i \le S_i^5$ for all large t. Set $A_5 = A_5(5)$ before choosing d. From (5.2),

$$S_i \le 6d\,t\sqrt{\log_2 t} + O(t), \qquad g(S_i) \le (6d + o(1))t.$$

Choose $6A_5 d < 1/4$. Polynomially many deterministic queries and the polynomial-size guesses then give a simulation below $2^{t/2}$ time, contradicting the hierarchy. Thus infinitely many lengths require more than $dN\sqrt{\log_2 N}$ wires. This is the two-stage Bathie–Williams proof instantiated at the present scale.

5.3 Every sufficiently large length

Assume H_{TC}. Let n denote the input length of the refuter, and choose

$$T(n) = 2^{\lceil n^{1/K} \rceil}$$

for a fixed sufficiently large integer K. It is time constructible and satisfies the refuter's range for all large n; finitely many smaller values can be adjusted. At the sampled lengths $n = t^K$, write $q(t) = t + O(\log t)$ for a fixed nondecreasing bound on PCP proof indices and random bits. The two-stage simulator A_{PCP} just described now runs on arbitrary $z \in \{0,1\}^n$, using the PCP for $A^T_{\mathrm{FS}}(z)$, with initial PCP soundness n^{-10}. It guesses the two normalized threshold circuits F, T, each capped by $B_d(q)$. The number of guessed bits is polynomial in t, with fixed degree: the two circuit descriptions have that property even though generating all PCPP constraints can take a larger polynomial time. Choose K greater than this degree. There are at most $n/10$ guessed bits for all large n.

The same error analysis shows $A_{\mathrm{PCP}}(z) \le A^T_{\mathrm{FS}}(z)$. Its time is

$$\mathrm{poly}(n) + \mathrm{poly}(t)2^{(6A_5 d + o(1))t} = o(2^t) \quad \text{if } 6A_5 d < 1.$$

The promise holds on every sufficiently large input length, with $t = \lceil n^{1/K} \rceil$. The refuter therefore produces, for each large sampled $n = t^K$, a z_t satisfying

$$A^T_{\mathrm{FS}}(z_t) = 1, \qquad A_{\mathrm{PCP}}(z_t) = 0. \tag{5.3}$$

It remains to encode both proof-bit functions at the *same* length. Let $h_0(t) = O(\log t)$ be a fixed canonical header length and $\ell(t) = O(\log t)$ the uniformly padded proof-bit index length. Use nondecreasing versions and define

$$P(t) = h_0(t) + 1 + q(t) + \ell(t), \qquad t(N) = \max\{t : P(t) \le N\}. \tag{5.4}$$

One may take $q(t) = t + C\lceil \log_2(t+2) \rceil$ and analogous ceiling-log bounds for the other lengths. Thus $P(t+1) - P(t) = O(\log t)$ and $N = q(t) + O(\log t)$ for $t = t(N)$. Use fields $(\mathrm{header}, b, x, j, \mathrm{padding})$, with $|x| = q(t)$ and $|j| = \ell(t)$. Invalid headers give zero. The $b = 0$ branch outputs the canonical fully accepted PCP proof bit $W_1(t, x)$ for z_t and ignores j; the $b = 1$ branch uses the canonical budgeted F_t and outputs $W_2(t, x, j)$ as above. Both branches ignore padding. The language is in E^{NP}: the refuter takes $2^{O(t)}$ time with SAT queries, as do the explicitly expanded canonical searches, and $t \le N$.

Suppose a circuit of at most $dN\sqrt{\log_2 N}$ wires computes this language at one sufficiently large N. Fix the correct header, padding, and branch bit. For W_1 also fix j. The resulting circuits have at most

$$dN\sqrt{\log_2 N} \le B_d(q(t))$$

wires for all large N, by the factor-two slack in (5.1). The W_1 circuit makes a fully accepted F_t exist; this ensures that the second is a universal PCPP proof circuit for that canonical F_t. Together they make $A_{\mathrm{PCP}}(z_t) = 1$, contradicting (5.3). This works separately at every sufficiently large N, proving the worst-case assertion of Theorem 1.2. Any sufficiently small positive rational d with $6A_5(5)d < 1$ is admissible.

5.4 Constant-advantage average-case hardness

Assume H_{TC}. We give the polynomial-parameter version of the proof in [4, Section 6], including the per-term size needed here. Let the two-query PCPP in [4, Lemma 3.11] have constants $0 < s_{\mathrm{p}} < c_{\mathrm{p}} < 1$ and gap $\Delta = c_{\mathrm{p}} - s_{\mathrm{p}}$. Its code bits are parities. Its constraint count is a power of two, and its proof constructor is polynomial time. Fix a rational

$$0 < \delta_0 \le \Delta^2/10^6. \tag{5.5}$$

Choose integer exponents K_1, K_2, K_3 as specified below, and then choose the hierarchy's K larger than the fixed degree of all guesses.

The cheating machine A_{PCPP} obtains the succinct PCP for $A^T_{\mathrm{FS}}(z)$ and guesses an *ordinary fan-in-two* circuit C_1 of size t^{K_1} as its oracle. The composed verifier is an ordinary polynomial-size circuit. Apply the two-query PCPP, with variables $Y \cup Z$, $m = \mathrm{poly}(t)$ constraints, and code word $Y = \mathrm{Enc}(x)$. Enlarge fixed polynomial bounds on proof lengths and constraint count uniformly across all candidate C_1; the native power-of-two constraint convention makes binary indexing uniform. To reach a common larger power of two, repeat every original constraint equally many times, preserving all satisfaction averages. The machine guesses, for each proof variable, a rational linear sum of at most $M = t^{K_2}$ threshold circuits, each with at most s_1 wires, and total coefficient absolute value at most t^{K_2}. Coefficients have a fixed $O(\log t)$-bit bound large enough for the constructions below. These guesses are not assumed to take values in $[0, 1]$.

For a constraint i, let T_{i1}, T_{i2} be its two (unsigned) variable functions. Let F_i be its OR polynomial after replacing a positive literal by T and a negative one by $1 - T$. Define

$$P_{ij} = \begin{cases} T_{ij}^2(1 - T_{ij})^2, & T_{ij} \text{ corresponds to a } Z \text{ variable,} \\ (\mathrm{Enc}_s - T_{ij})^2, & T_{ij} \text{ corresponds to } Y_s, \end{cases} \qquad Q_{ij} = T_{ij}^2. \tag{5.6}$$

Estimate expectations of all these polynomials by expanding products of at most four sum terms. Each expectation is a linear combination of at most polynomially many expectations of ANDs of at most four threshold circuits, together with an $O(q)$-wire parity when required. Every CAPP query has

$$w \le 4s_1 + O(q). \tag{5.7}$$

The parity is computed as part of the circuit; no enumeration of its input support is needed. Taking per-product error t^{-K_3} gives total expectation error $O(t^{4K_2 - K_3}) = o(1)$ when $K_3 > 4K_2$. Increase K_3 so all errors below are eventually at most $\delta_0/10$.

Reject unless the estimated average of P_{ij} is at most $2\delta_0$ and every estimated Q_{ij} is at most $1 + \delta_0$. After this validity test, accept precisely when

$$\frac{1}{m} \sum_i \widetilde{\mathbb{E}}_x F_i \ge c_{\mathrm{p}} - \Delta/2. \tag{5.8}$$

These are the tests of [4, Section 6.2.2]; their requisite estimates involve only (5.7).

Here are the validity estimates explicitly. If the guessed functions are $[0, 1]$-valued and their average ℓ_1 distance, over occurrences, from a Boolean PCPP proof is at most δ_0, then the true average of P is at most δ_0 and every Q is at most one. Thus they pass. Conversely, passing implies a true average $P \le 3\delta_0$ and $\mathbb{E}Q_{ij} \le 1 + 2\delta_0$. Round every Z function to its nearest Boolean value, and set $Y = \mathrm{Enc}(x)$. For real u,

$$\min\{u^2, (1 - u)^2\} \le 4u^2(1 - u)^2.$$

Hence, by Cauchy–Schwarz, the average L_2 distance of the guessed occurrences from the rounded proof is at most $\sqrt{12\delta_0}$. For each OR polynomial, telescoping its two arguments,

using the Boolean rounded argument and $\|T_{ij}\|_2 \le \sqrt{1+2\delta_0}$, gives

$$\|F_i - \widehat{F}_i\|_1 \le 7\,\mathbb{E}_{j\in\{1,2\}}\|T_{ij} - \widehat{T}_{ij}\|_2.$$

For a negated literal use $\|1 - T_{ij}\|_2 \le 1 + \sqrt{1+2\delta_0}$ in the same telescoping; this gives the constant 7. If $A^T_{\mathrm{FS}}(z) = 0$, the initial PCP accepts on at most n^{-10} of the x's, so every Boolean PCPP proof has average satisfaction at most $s_{\mathrm{p}} + o(1)$. The guessed proof's average is at most

$$s_{\mathrm{p}} + 7\sqrt{12\delta_0} + o(1) < c_{\mathrm{p}} - \Delta/2.$$

Including estimation error preserves the strict inequality. Therefore $A_{\mathrm{PCPP}}(z) \le A^T_{\mathrm{FS}}(z)$ for all large inputs.

Now fix $\gamma \in (0, 1/2)$, and choose the least $k \ge 1$ with

$$\varepsilon_k = (1-\delta_0)^{k-1}(1/2 - \delta_0) \le \gamma/2. \tag{5.9}$$

Let c_{STV} be a fixed integer upper bound on the input-length blow-up in the ordinary-circuit amplification [4, Lemma 3.9], originating in [16], and put

$$k' = \max\{k, c_{\mathrm{STV}}\}. \tag{5.10}$$

Take $s_1 = B_{dk}(q)$. Fix $a_* \ge \max\{5, K_3\}$ large enough for all query descriptions and precisions, and write $A_5 = A_5(a_*)$. Choose a small positive rational d such that

$$8A_5 dk < 1. \tag{5.11}$$

Indeed, (5.7) has $4s_1 \le 8dkq\sqrt{\log_2 q}$, so the machine's time is $\mathrm{poly}(t)2^{(8A_5dk+o(1))t} = o(2^t)$. The fixed exponent a_* is chosen to cover the required inverse-polynomial error; the constant A_5 need not equal $A_5(5)$ from the worst-case argument. All guesses have polynomial length in t, so a sufficiently large K gives the refuter promise. Obtain z_t with $A^T_{\mathrm{FS}}(z_t) = 1$ and $A_{\mathrm{PCPP}}(z_t) = 0$.

There are two cases, decidable in $2^{O(t)}$ time with SAT queries by expanding all PCP random strings. If no C_1 of size t^{K_1} gives a fully accepted oracle, the canonical fully accepted proof's truth table has ordinary circuit complexity greater than t^{K_1}. The amplification in [4, Lemma 3.9] constructs, in $2^{O(q)}$ time from that truth table, a function g on at most $c_{\mathrm{STV}}q$ inputs that is $(1/2 + t^{-1})$-hard for ordinary size t^3, provided K_1 exceeds three times the fixed amplification exponent. The Boolean simulation of threshold circuits above makes g hard for the required near-linear threshold wire budget as well.

In the other case take the lexicographically least successful ordinary C_1 and a canonical Boolean PCPP proof $\widehat{T}$, using the proof constructor separately for each x. Set

$$f(i,j,x) = \widehat{T}_{ij}(x), \qquad n' = q + \log_2 m + 1 = t + O(\log t).$$

This function is δ_0-far in ℓ_1 from every $[0,1]$-valued linear sum with at most $M/(2m)$ terms, each of at most s_1 wires, and coefficient absolute sum at most $t^{K_2}/(2m)$. To see this, suppose D is a closer sum and define the guessed function for each proof variable X by averaging $D(i,j,x)$ over occurrences of X. Empty occurrence lists can be assigned zero. Each resulting function has at most M terms, no larger per-term wire bound, and no larger coefficient absolute sum than the allowed cap. For $v_i \in [0,1]$ and $b \in \{0,1\}$,

$$\left|\frac{1}{h}\sum_i v_i - b\right| = \frac{1}{h}\sum_i |v_i - b|.$$

Apply this equality for each x and occurrence list: the average ℓ_1 distance to $\widehat{T}$ remains at most δ_0. The validity test passes, and the OR polynomials, Lipschitz on $[0,1]^2$, lose at

most $2\delta_0$ average satisfaction relative to the canonical proof. They therefore satisfy (5.8), contradicting $A_{\mathrm{PCPP}}(z_t) = 0$.

Apply the *proof of Lemma 3.8 in Appendix A* of [4], rather than only its complexity statement. For completeness, its reconstruction has the following form. Given a circuit approximating $f^{\oplus k}$, either fix one block and recurse on $k-1$, possibly negating the circuit, or sample $\ell = O(n'/(\delta_0\varepsilon_k)^2)$ fixed values of the other blocks. A Chernoff bound and a union bound over the $2^{n'}$ remaining inputs make the empirical conditional agreement uniformly accurate. An affine rescaling of that empirical mean produces a $[0,1]$-valued linear sum approximating f within δ_0 in ℓ_1. Every term is a projection, possibly negated, of the original approximator. In particular, each term retains its original wire bound; there are $O_\gamma(t)$ terms, and the sum of coefficient absolute values is $O_\gamma(1)$, or the weaker sufficient $O_\gamma(t)$. All coefficients are rational with $O_\gamma(\log t)$ bits because δ_0 and ε_k are rational. Increasing the fixed K_2 makes these sums fit the preceding caps. Thus $f^{\oplus k}$ cannot have agreement $1/2 + \varepsilon_k$ with the threshold approximators under consideration.

Finally use a nondecreasing payload bound

$$P_\gamma(t) = h_0(t) + \max\{k(q(t) + \log_2 m(t) + 1),\ c_{\mathrm{STV}} q(t)\} = (k' + o(1))t,$$

where polynomial padding of the PCPP supplies a common dyadic $m(t)$ bound. For each input length N choose the largest t with $P_\gamma(t) \le N$, and read the active function input from the leading payload bits, ignoring all other bits. The length N determines t; no header is validated and every word of length N is used. In Case 1, output g; in Case 2, output $f^{\oplus k}$. Canonical searches, case selection, and truth-table amplification all run in $2^{O(t)}$ time with an NP oracle, so this defines $L_\gamma \in \mathrm{E}^{\mathrm{NP}}$. Set $c_\gamma = dk/k'$. A length-N approximator of at most $c_\gamma N\sqrt{\log_2 N}$ wires has

$$s_0 \le (dk + o(1))t\sqrt{\log_2 t} < B_{dk}(q) = s_1$$

wires for all large t. Averaging over padding gives a fixing with the same agreement. Both cases yield a contradiction to agreement greater than $1/2 + \gamma$. One can take

$$c_\gamma = \Theta\left(\frac{1}{A_5 k'}\right) = \Theta\left(\frac{\delta_0}{A_5 \log(1/\gamma)}\right) \quad \text{as } \gamma \downarrow 0.$$

This proves the average-case assertion of Theorem 1.2. The proof uses inverse-polynomial CAPP precision throughout.

Part II
Sparse connectivity

6 Sparse determinants with linearly many Boolean wires

The determinant will be a nonnegative connectivity weight after its permutation terms have been summed. The required tensor representation must therefore preserve permutation signs as well as matrix entries. We give the sign construction explicitly; the inverse-pair accounting is related to the determinant method in [3, Section 4].

6.1 A sparse permutation-sign gadget

Let $E \subseteq [N] \times [N]$ be the potential nonzero positions of an $N \times N$ matrix, let $M = |E|$, and set

$$T_{\mathrm{det}} = N + M + 2.$$

Position bits z_{ij} select one entry per row and column. A legal selection is a permutation σ supported on E. We construct its sign with $O(T_{\mathrm{det}})$ edges and maximum degree $O(T_{\mathrm{det}}^{2/3})$. The following construction is a structural identity used under our standing assumption H_{TC}.

Put $B = \max\{1, \lfloor T_{\mathrm{det}}^{2/3} \rfloor\}$. Rows having more than B potential positions are heavy; each is a singleton block. Split every consecutive run of light rows greedily, preserving row order, so that the total number of potential positions in a block is at most B. Columns are treated independently in their own order. A zero potential row or column makes the determinant identically zero and can be handled directly.

If there are a row blocks and b' column blocks, then

$$a, b' = O(M/B + 1), \qquad ab' = O(T_{\mathrm{det}}^{2/3}). \tag{6.1}$$

Indeed there are at most M/B heavy rows. In one light run, any two consecutive greedy blocks have total load greater than B, so its block count is at most twice the load divided by B, plus one. There are at most one more light runs than heavy rows. In particular $a \le 4M/B + 1$, and the column bound is identical. In a cell $E_{uv} = E \cap (R_u \times C_v)$, a light side bounds its size by B; if both sides are heavy singletons it contains at most one position.

For a legal permutation write $r(i)$ for its row-block index, $c(j)$ for its column-block index, and $n_{uv} = |\{i \in R_u : \sigma(i) \in C_v\}|$. Define

$$\begin{aligned}
I_R &= |\{i < i' : r(i) = r(i'),\ c(\sigma(i)) > c(\sigma(i'))\}|,\\
I_C &= |\{j < j' : c(j) = c(j'),\ r(\sigma^{-1}(j)) > r(\sigma^{-1}(j'))\}|,\\
I_{uv} &= |\{i < i' : i, i' \in R_u,\ \sigma(i), \sigma(i') \in C_v,\ \sigma(i) > \sigma(i')\}|,\\
I_\times &= \sum_{u<u',\ v>v'} n_{uv} n_{u'v'}.
\end{aligned}$$

The exact integer identity is

$$\mathrm{inv}(\sigma) = I_R + I_C + \sum_{u,v} I_{uv} + I_\times. \tag{6.2}$$

To prove it, classify a pair $i < i'$ by whether its two row blocks are equal and whether its two image column blocks are equal. If both are equal, it is counted in I_{uv}. If only the row blocks are equal, order preservation makes a column inversion equivalent to a column-block inversion, counted in I_R. If only the column blocks are equal, let $j < j'$ be the two image columns; an inversion is exactly a reversal of the corresponding row-block order, counted in I_C. If neither is equal, inversions occur precisely in cells with $u < u'$ and $v > v'$, giving $I_\times$. The four classes are disjoint.

Use constant-state exact-one chains to check row and column selection. Each light row block reads its at-most-B position bits and contributes $(-1)^{I_R}$ for that block; a heavy singleton contributes one. Column blocks similarly contribute $(-1)^{I_C}$. Each cell reads its position bits, contributes $(-1)^{I_{uv}}$, and outputs one parity bit

$$p_{uv} = \sum_{(i,j) \in E_{uv}} z_{ij} \pmod 2.$$

It can also reject repeated selected rows or columns in the cell. Empty cells output a forced zero bit. A single factor reads the ab' parity bits and contributes

$$(-1)^{\sum_{u<u',\ v>v'} p_{uv} p_{u'v'}}. \tag{6.3}$$

This gives $(-1)^{I_\times}$. Its degree is bounded by (6.1). Every selected-position bit occurs in only a constant number of checks. Total port count is $O(N + M + ab') = O(T_{\mathrm{det}})$; maximum

degree is $O(B + ab') = O(T_{\det}^{2/3})$. Each entry is computed by polynomially many arithmetic and pair-comparison operations on its ports. Exact-one states and cell parities are unique on each legal selection. Equality trees and subdivision preserve these bounds.

For each position multiply the factor

$$z_{ij} = 0 : \ 1, \qquad z_{ij} = 1 : \ A_{ij}.$$

Summing over the position bits and the unique states gives precisely

$$\sum_{\sigma \text{ supported on } E} \operatorname{sgn}(\sigma) \prod_i A_{i,\sigma(i)} = \det A. \tag{6.4}$$

The A_{ij} may depend on external Boolean variables. If those variables and the total scopes of their conditions and numerical entry factors are $O(T_{\det})$, the combined representation still has linear edges. Its maximum degree is the larger of the sign-gadget degree and the external degree. This is a representation of a determinant inside a larger sum over external choices.

6.2 Marker Laplacians and positivity

Fix a simple undirected graph $G = (V, E_G)$, $U \subseteq V$, $F \subseteq E_G[U]$, and markers $M_0 \subseteq U$. Define

$$A_{vv} = d_F(v) + [v \in M_0] + [v \notin U], \qquad A_{vw} = -[vw \in F] \quad (v \neq w). \tag{6.5}$$

The unselected vertices form an identity block. On U, the matrix is the Laplacian of the graph H^+ obtained from $H = (U, F)$ by adjoining one root ρ and the edges ρv for $v \in M_0$, with the root row and column deleted.

For completeness, orient the edges of H^+ arbitrarily, and let B_ρ be its incidence matrix with the root row removed. Then $A|_U = B_\rho B_\rho^T$, and Cauchy–Binet gives

$$\det A = \sum_{\substack{Q \subseteq E(H^+) \\ |Q| = |U|}} \det((B_\rho)_Q)^2 = \tau(H^+). \tag{6.6}$$

A cyclic edge set has dependent incidence columns. An acyclic set of $|U|$ edges on $|U| + 1$ vertices is a spanning tree. Its incidence minor has determinant ± 1, by repeatedly expanding at a nonroot leaf. Thus every tree contributes one and every other set contributes zero. Empty U gives determinant one. In particular,

$$\det A \geq 0, \qquad \det A > 0 \iff \text{every component of } (U, F) \text{ meets } M_0. \tag{6.7}$$

If $M_0 = \{r\}$ with $r \in U$, positivity is connectivity of (U, F), and $\det A = \tau(U, F)$ because the root's single incident edge is a bridge.

The potential positions in (6.5) number at most $n + 2m$. For local evaluation, compute every $d_F(v)$ by an $O(\deg_G(v) + 1)$-gate Boolean population counter. One construction repeatedly uses constant-size full adders to reduce triples in each binary column: each full adder reduces the total number of live bits by one, so $O(d)$ suffice for d input bits, followed by addition of two $O(\log(d + 2))$-bit words. Its internal states are functional. All degree circuits together have $O(n + m)$ wires and gates. Their output-bit count is $O(n + m)$ as well. A diagonal numerical factor reads only its degree output, marker bit, selected-vertex bit, and position bit; an off-diagonal factor reads a position bit and an edge-selection bit. Total scope length stays $O(n + m)$, including at vertices of high graph degree.

6.3 Exact statistics with long binary coefficients

The applications also use a fixed number of statistics $A_j(y) = \sum_i a_{ji} y_i$ on $p = O(S)$ object bits, with nonnegative coefficients of at most $H = \lceil S^{9/10} \rceil$ bits. Any fixed polynomial-time relation on these statistics, including a threshold, can be added using $O(S)$ wires, degree $O(S^{9/10})$, and a unique auxiliary extension.

Indeed, group the object bits in blocks of $B = \max\{1, \lfloor S^{9/10} \rfloor\}$ bits. There are $h = O(S^{1/10} + 1)$ blocks. Every partial or total sum fits in

$$q = H + \lceil \log_2(p+1) \rceil + 1 = O(S^{9/10})$$

bits. A block factor reads its original bits and checks its output partial sums. For each statistic, a binary tree of addition factors combines these outputs. Each addition factor reads three q-bit words, rejects overflow, and enforces exact addition; on genuine inputs there is no overflow. The root relation reads only the fixed number of total sums. The total number of state wires is $O(hq) = O(S)$, whereas the maximum degree is $O(S^{9/10})$. The use of an addition tree, rather than one factor reading every block sum, is essential to this degree bound. Local entry arithmetic is polynomial in $S + L$.

For coefficients bounded in value by $S^{O(1)}$, a smaller-block variant also works: take blocks of $S^{2/3}$ inputs, transmit $O(\log S)$-bit partial sums, and let a terminal factor read all of them. Its degree is $O(S^{1/3} \log S)$, and the block degree is $O(S^{2/3})$. Both encodings are compatible with the determinant representation. The larger-degree one uses the constant rescaling (2.2) before invoking H_{TC}.

7 Undirected connectivity: exact sums, optimization, and witnesses

We prove Theorem 1.3, always assuming H_{TC}. Let $S_G = n + m + 2$. The determinant scale in Section 6 is

$$T_{\det} = n + (n + 2m) + 2 = 2n + 2m + 2 = O(S_G).$$

These are distinct parameters; the precision below is measured in S_G.

7.1 Local candidate systems

Write u_v for selected-vertex bits and x_e for selected-edge bits. Requiring selected-edge endpoints to be selected is a constant scope condition. Incidence degree conditions, including exact degrees zero, one, and two, have $O(\deg_G(v) + 1)$-wire functional chains: pass the saturated count $\min\{3, \text{count so far}\}$ in two bits. Domination is checked by an OR chain on each closed neighborhood, with a total of $O(n + m)$ gates. Vertex cover has the binary constraints $u_v \vee u_w$ for every edge vw. In induced systems introduce the unique bit $x_{vw} = u_v u_w$ by an AND relation. All such systems have $O(S_G)$ original and auxiliary bits and total scopes $O(S_G)$. Global cardinalities and cost relations are the fixed number of statistics from Section 6.3.

For each candidate choose the matrix (6.5) and combine its sparse determinant gadget with these constraints. The contraction is exactly

$$Z = \sum_{y \text{ satisfying the local and target conditions}} \det A(y). \tag{7.1}$$

The sum over permutation bits is taken inside the sum over y; all other auxiliary states have unique extensions. Table 2 lists the candidates and the determinant value on the objects that survive the filter. Each single-root row explicitly requires $r \in U$.

Problem	Candidate constraints and markers	Surviving weight
HC / Hamiltonian TSP	$U = V$, all selected degrees two; $M_0 = \{r\}$	n for a Hamiltonian edge set
Hamiltonian path	$U = V$, degrees one at distinct s, t, two elsewhere; $M_0 = \{s\}$	1 for a spanning s–t path
Simple path	$s, t \in U$, degree one at s, t, two at other selected vertices, zero outside; $M_0 = \{s\}$	1 for an s–t path on U
Steiner tree	Terminals in U, $F \subseteq E_G[U]$, $\lvert F\rvert = \lvert U\rvert - 1$; $M_0 = \{r\}$ for a fixed terminal	1 for a tree (U, F)
Connected vertex cover	Cover constraints, $F = E_G[U]$, $M_0 = \{r\}$	$\tau(G[U]) > 0$
Connected domination	Domination constraints, $F = E_G[U]$, $M_0 = \{r\}$	$\tau(G[U]) > 0$
Feedback vertex set	$F = E_G[U]$, $M_0 \subseteq U$, $\lvert F\rvert + \lvert M_0\rvert = \lvert U\rvert$	1 for each forest with one marker per component

Table 2: In every row the determinant is zero on candidates failing the required connectivity condition. Costs and a specified path length can be added as target relations.

Here is the verification of each row. A simple graph of degree two at every vertex is a disjoint union of cycles. A single marker allows only the component containing that marker. A surviving cycle on n vertices has exactly n spanning trees, obtained by deleting one edge. Consequently

$$Z = n \,\#\mathrm{HC}. \tag{7.2}$$

No orientation or starting-vertex multiplicity is present: the objects are undirected edge sets. The matrix remains unweighted even when cost constraints are imposed on these edge sets.

For path candidates, the two odd-degree endpoints must lie in the same path component, and every other component is a cycle. The single marker eliminates those cycles; the remaining path has tree number one. For a specified length ℓ, impose $|U| = \ell + 1$. For Steiner candidates, connectivity and $|F| = |U| - 1$ make the chosen subgraph a tree. The counted objects are all tree subgraphs containing the terminal set, including ones with nonterminal leaves. For connected covers and dominating sets, the determinant is their induced subgraph's positive tree count, so positivity of (7.1) is exactly existence.

For FVS, append a root adjacent to every marker. When the determinant is positive, this augmented graph is connected, and the imposed equation makes its number of edges equal to its vertex count minus one. It is therefore a tree and the determinant equals one. Removing its root leaves the induced forest $G[U]$, with exactly one marker per component. Conversely every induced forest has such marker choices, all of weight one. Thus $Z > 0$ decides whether a forest of the specified deletion cost exists. Different marker choices give a positive multiplicity for the same deletion set.

7.2 Precision and exact readback

All matrices in (7.1) have nonnegative determinant by (6.6). Each augmented graph has at most $m + n$ edges, so its tree count is at most 2^{m+n}. The numbers of candidates (U, F), (U, M_0), or U are at most 2^{n+m}, 3^n, and 2^n, respectively. Uniformly,

$$0 \le Z \le 2^{\max\{2n+2m,\, 3n+m\}} < 2^{4S_G}. \tag{7.3}$$

Take

$$P_0 = 4S_G + 1, \qquad \ell = 1. \tag{7.4}$$

The local signed entries, including -1, are legitimate residues in $\mathbb{Z}/2^{P_0}\mathbb{Z}$. Their intermediate products may cancel; the final sum is the bounded nonnegative integer in (7.3). Its least nonnegative residue is therefore exactly Z. For HC divide that integer by n, as in (7.2). Even when n is even, no inverse modulo a power of two is required.

The local candidate constraints, degree counters, and numerical matrix factors have linear total scope length. The sparse sign gadget has degree $O(S_G^{2/3})$; adding long cost constraints raises the degree to $O(S_G^{9/10})$ and keeps $O(S_G)$ wires. Choose $s = \lceil KS_G \rceil$ with a fixed sufficiently large K, and subdivide edges for bipartiteness. Precision (7.4), vertex count, description length, and polynomial-time local evaluation then satisfy every part of H_{TC}. One query takes

$$\mathrm{poly}(S_G + L)2^{O(g(S_G))} \tag{7.5}$$

time and polynomial space.

7.3 Optimization, witnesses, and boundary conventions

Nonnegative edge and vertex costs are attached to the *object* bits, not to the spanning trees counted by the determinant. For FVS the objective is the cost of $V \setminus U$, implemented with complement bits. Use a cost-at-most predicate for minimization and a cost-at-least predicate for maximization. The sum of the input costs has $O(n^{9/10} + \log n)$ bits on a sparse graph, and binary search needs only polynomially many queries. For longest path one may maximize the number of selected vertices, or an explicitly given nonnegative selected-edge/vertex weight. The long-coefficient gadget handles these exact target relations without enumerating cost values.

If a target has $Z > 0$, fix its object bits successively. For a bit y_i we have $Z = Z_{y_i=0} + Z_{y_i=1}$, with both summands nonnegative. Keep any positive child. Apply this only to object bits (and, for FVS, marker bits), retaining the determinant's permutation sum. At the end the surviving object satisfies the target and its graph property. Direct polynomial-time graph and cost checks verify the returned witness. At most a linear number of additional queries is needed.

Enumerating possible roots and endpoint pairs costs only a polynomial factor. For exact counting of unspecified-endpoint undirected paths, use $s < t$ to count each nontrivial edge set once. Single-vertex paths are counted separately. For prescribed $s = t$, only the zero-length single-vertex path is included. Simple undirected cycles require at least three vertices. An empty terminal set allows the empty Steiner tree of cost zero in optimization. For counting all nonempty Steiner tree subgraphs with no terminals, choose the unique root $r = \min U$ and force smaller vertices outside U; add the empty tree only if that convention is requested. With a nonempty terminal set, fix one terminal as the root, so that no root multiplicity is introduced. Single-root trees are included. Empty-cover and empty-graph conventions are checked directly; all nonempty connected sets are covered by the root enumeration.

The number of queries, including roots, endpoints, binary search, and self-reduction, is polynomial. Their workspace is reused. Since $m \le cn$, $g(S_G) = O_c(g(n))$, proving Theorem 1.3.

8 Directed graphs and sparse-base routing

8.1 The directed marker identity

Assume H_{TC}. Let a loopless digraph have selected vertices U, selected arcs F with endpoints in U, and markers $M_0 \subseteq U$. Replace (6.5) by the outgoing Laplacian

$$A^+_{vv} = d^+_F(v) + [v \in M_0] + [v \notin U], \qquad A^+_{vw} = -[v \to w \in F] \quad (v \neq w). \tag{8.1}$$

Its determinant counts spanning arborescences directed toward the new root ρ, after adding $v \to \rho$ for every marker v. Here every selected vertex has one outgoing tree arc and every orbit reaches ρ.

We verify this identity directly. Remove the identity block outside U. Expand each diagonal entry as a choice of one outgoing selected arc or an available root arc. Fix all these choices, a function $f : U \to U \cup \{\rho\}$. Its nonroot directed cycles are disjoint. A nontrivial permutation cycle in a determinant term must be one of these cycles: its off-diagonal entries require the corresponding choices of f. Conversely any subset of these cycles may be used, with every other vertex a diagonal fixed point. A length-d cycle has permutation sign $(-1)^{d-1}$ and d negative entries, so contributes -1. The total for this f is

$$\sum_{Q \subseteq \mathrm{cycles}(f)} (-1)^{|Q|} = (1-1)^{|\,\mathrm{cycles}(f)|}.$$

It is one exactly when every orbit reaches the root. Thus $\det A^+ \geq 0$, and it is positive exactly when every selected vertex can reach a marker along selected arcs. For the converse existence claim, choose for each vertex an arc decreasing its shortest distance to a marker or to the new root. Empty U has determinant one. The exclusion of loops keeps diagonal choices separate from permutation cycles.

The potential matrix positions number at most $n + m$, and outgoing degree counters and entry scopes are $O(n + m)$. Combining this identity with the sparse sign gadget obeys the same wire, degree, entry-cost, and ring bounds as before.

8.2 Directed applications

Proof of Theorem 1.4. For directed HC, require indegree and outdegree one at every vertex and fix one marker r. Candidates are disjoint directed cycles. A spanning cycle has exactly one rooted arborescence in the augmented graph: replace r's outgoing cycle arc by its root arc, and keep the other arcs. Any additional cycle misses the marker and has determinant zero. The final sum is therefore the directed HC count itself; it is not divided by n.

For a directed s–t path, $s \neq t$, require selected indegree/outdegree $(0, 1)$ at s, $(1, 0)$ at t, $(1, 1)$ at other selected vertices, and $(0, 0)$ outside U. Use t as the sole marker. The path contributes one, and any extra directed cycle makes the determinant zero. This proves exact counting with a prescribed length and longest-path optimization. Unspecified endpoints are enumerated as *ordered* pairs. A single-vertex path is treated separately. A two-vertex directed cycle with opposite arcs is a valid HC under these conventions. Empty graphs and the one-vertex loopless case are checked directly.

For directed Steiner tree toward a root r, require $r \in U$, outdegree zero at r, outdegree one at each other selected vertex, and inclusion of all terminals. With marker r the determinant is one precisely on arborescences toward r. A feasible rooted-reachability subgraph contains a suitable arborescence, obtained by selecting arcs that decrease distance to the root; extra branches not needed for the terminals can be deleted. With nonnegative costs this restriction preserves the optimum. Reverse the input for the usual version with paths from the root to the terminals. The same network exactly counts arborescent subgraphs of the specified direction. Even when terminals consist only of the root, the counting problem includes all such subgraphs; only the optimum can immediately be the single root.

For a strongly connected selected subgraph, fix $r \in U$ and multiply two determinant gadgets on the *same* selected-arc bits: one for F and one for its reverse. The second matrix's diagonal uses the original indegrees; it is recomputed rather than obtained by merely transposing the original matrix. The product is positive exactly when every selected vertex can reach r and can be reached from r. This is strong connectivity. The product stays inside the common sum over candidates. It is not the product of two separately contracted sums. For

a spanning subgraph use $U = V$. For a terminal version use $T \subseteq U$ and a fixed $r \in T$ when T is nonempty. For an empty terminal set use the requested empty-subgraph convention, or enumerate roots to cover nonempty solutions.

Each arborescence is a subset of at most $m + n$ potential augmented arcs. At most two determinant factors and at most 2^{n+m} candidates give the uniform bound

$$0 \le Z \le 2^{3(n+m)} < 2^{4(n+m+2)}.$$

Use $P_0 = 4(n + m + 2) + 1$, $\ell = 1$, and a fixed multiple of $n + m + 2$ for the H_{TC} scale. Costs use the exact statistics gadget; objective and witness bits are recovered by the same positive-sum self-reduction as in the undirected case. All query counts are polynomial. This proves the theorem. □

8.3 Closed walks, required edges, and metric TSP

Proof of Theorem 1.5. A nonempty closed walk determines a connected even-degree edge multiset. If an edge has multiplicity at least three, delete two of its copies. Its support remains unchanged, degrees stay even, and required-edge and terminal constraints remain satisfied. The cost does not increase. Repeating gives an optimum with multiplicities

$$\mu_e \in \{0, 1, 2\}. \tag{8.2}$$

Conversely a connected even-degree multigraph has an Euler tour: follow unused edges to a closed trail, and splice in further closed trails wherever unused edges remain. Connectivity and parity ensure this process exhausts every edge. A selected single vertex without edges represents its zero-length closed walk.

Encode $\mu_e = a_e + 2b_e$ with Boolean bits and forbid $a_e = b_e = 1$. Define its support $f_e = a_e \vee b_e$ uniquely. Required edges have $f_e = 1$. Select U containing the terminals and every support endpoint; the single-marker determinant on the support graph $(U, \{e : f_e = 1\})$ enforces connectivity. Extra isolated selected vertices are thereby eliminated except for the valid singleton case. For nonempty requirements fix $r \in T \cup V(R)$ as the marker and require $r \in U$. If a depot is specified, include it in T. Empty requirements allow an empty walk of cost zero and are handled directly.

A parity chain at vertex v requires

$$\sum_{e \ni v} a_e \equiv 0 \pmod 2,$$

which is exactly the parity of its multidegree. The chains and the multiplicity/support relations together have $O(n + m)$ wires. The cost is the exact statistic

$$\sum_e w_e a_e + 2 w_e b_e.$$

Its coefficient width is at most one more than the input width; this constant change is covered by the fixed-scale long-coefficient gadget. The determinant counts trees of the *simple support* graph, as a positive feasibility weight.

There are at most $2^n 3^m$ candidates (U, μ), and at most 2^m support spanning trees for a single marker. Thus

$$0 \le Z \le 2^n 6^m < 2^{4(n+m+2)}.$$

The same precision and H_{TC} scale apply. Binary search on cost, followed by fixing a_e, b_e, u_v while preserving positive sums, recovers an optimal even connected multiset. Its Euler tour has at most $2m$ edge traversals. With a depot, rotate the tour to start and end there.

Finally suppose the base graph is connected, $T = V$, and R is empty. Expand any Hamiltonian tour of its shortest-path metric into base-graph shortest paths. This gives a closed spanning walk of the same cost. Conversely, visit vertices in their order of earliest appearance along a spanning walk and shortcut repeats in the metric. The triangle inequality gives a Hamiltonian metric tour of no greater cost. Therefore the optima are equal, and shortcutting an optimal returned walk gives an optimal metric tour. Zero costs can give a pseudometric; the same inequalities still hold. Distances are needed only for this polynomial-time output step; a dense metric closure is not part of the tensor network. The proof of (1.1) follows. □

9 Encoding boundaries and finite checks

9.1 Precision and representation boundaries

At $S = \Theta(n)$, the high-precision error in Theorem 1.1 is still much larger than 2^{-n}. Its absolute-error guarantee does not distinguish zero accepting inputs from one. Exact modes require their weight-cost budget or a supplied short factorization. A general polynomial-bit equality constraint on n input bits need not satisfy those exact budgets.

The sum in (4.1) is a sufficient condition for the constructions above; no necessity is claimed for circuits whose gates share realizations. The seed estimate used by the algorithm is $R = O_{a,c}(S^{9/10})$, measured in the scale S. Exact-only assignments have $R = 0$; hashed assignments use their pointwise error bounds for the precision comparison.

A ring calculation clarifies the cost of a particular congruence filter. In $R = (\mathbb{Z}/2^P\mathbb{Z})[V]/(V^\ell)$ let $\mathfrak{m} = (2, V)$. If units $\zeta_j \in R$ and coefficients $c_j \in R$ satisfy

$$\sum_j c_j \zeta_j^u = 2^e[u = 0], \qquad 0 \le u < M, \quad e < P,$$

then $e \ge M - 1$. Every unit is one modulo $\mathfrak{m}$; the $(M-1)$st forward difference of the left side belongs to $\mathfrak{m}^{M-1}$, while that of the right side is $(-1)^{M-1}2^e$. The constant coefficient of $\mathfrak{m}^{M-1}$ is divisible by 2^{M-1}, forcing the bound. Filters of this specified unit-power form therefore consume precision proportional to their modulus range.

9.2 The fixed-width counting interface

The local condition "two selected incident edges at every vertex" recognizes a spanning 2-factor. Two disjoint triangles satisfy it and have no HC. Joining them by a bridge and leaving the bridge unselected gives the same example on a connected input. Rooted consistent cuts give exactly $2^{\mathrm{cc}(F)-1}$ extensions and filter disconnected 2-factors modulo two. Distinct Hamiltonian solutions in one weight layer can still cancel. The deterministic proofs use nonnegative determinant sums instead.

Known fixed-width CNF encodings used in this comparison do not by themselves derive these sparse connectivity bounds from H_{BB}. Numbering-based certificates with $\Theta(n \log n)$ Boolean variables give, even with fixed clause density, only the displayed bound $2^{O(n\sqrt{\log n})}$. Compiling weighted isolation sums into Boolean arithmetic has a similar logarithmic cost. The implicit sign and arithmetic factors are where the graph proofs use H_{TC}. Whether a smaller encoding or a different reduction derives these bounds from H_{BB} alone remains open in this analysis.

One H_{BB}-only example can be verified directly. On a sparse graph, exact chromatic number and an optimal coloring are computable, conditional on H_{BB}, in $\mathrm{poly}(n + L)2^{O(n \log\log n/\sqrt{\log n})}$ time and polynomial space. Take $D = \lceil \log_2(n + 2) \rceil^2$ and let H contain vertices of original degree greater than D. Then $|H| = O(n/\log^2 n)$.

Compute $\chi(G[H])$ by enumerating all $|H|^{|H|}$ color labelings. If it is at least $D+1$, an optimal coloring of H extends greedily to the remaining vertices and is optimal. Otherwise try $q \le D+1$ and every $q^{|H|}$ coloring of H. Represent each remaining color in $O(\log q)$ bits, reject codes at least q, and test each remaining or crossing edge with an $O(\log q)$-gate inequality circuit. The functional Tseitin 3-CNF has $O(n \log\log n)$ variables and clauses. Add one fixed-zero dummy bit per clause, plus twenty, to make density less than two while preserving the count. Apply H_{BB} with fixed $k = 3$ and density two; all enumeration factors are absorbed in the stated exponent. Fixing original color bits recovers an optimal extension. Empty graphs and the case $q = 1$ are checked directly. This is a routine composition of degree splitting, binary comparison, and H_{BB}.

9.3 Budget scales in lower-bound simulations

All H_{TC} calls in Section 5 go through Theorem 1.1, whose scale is the actual S of the query circuit; the proofs use only upper bounds on the resulting running time, so no lower bound on the size of a guessed circuit is needed.

For average-case amplification, the input length grows by k and the analysis-query exponent is proportional to ck. Condition (5.11) permits constant k at a fixed positive wire constant. Increasing CAPP precision alone does not remove that time constraint.

9.4 Classical sparse baselines

The sparse reductions for the usual ETH comparison have explicit linear budgets. Sparsification [9] gives a 3-CNF with v variables and $a = O(v)$ clauses. The standard vertex-cover construction has a variable edge and a clause triangle, with each clause vertex joined to its literal's variable vertex. Its budget is $v + 2a$; there are $2v + 3a$ vertices and at most $v + 6a$ edges. The forced one-per-variable and two-per-triangle choices make an unchosen clause vertex imply a true literal. The standard dominating-set construction has triangles (t_i, f_i, z_i) and a vertex for each clause joined to its literals, with budget v. The private z_i forces one selection per triangle; choosing z_i can be replaced by a literal. It has $3v + a$ vertices and at most $3v + 3a$ edges.

From vertex cover, make a new triangle on each retained edge by adding one vertex. Deleting a cover leaves a forest; a selected new vertex in an FVS can be replaced by an endpoint without creating a cycle. Adding a universal root and $k + 2$ private leaves gives connected vertex cover with budget $k + 1$. For a dominating-set instance use selectors a_u, elements b_v, edges $a_u b_v$ for $v \in N[u]$, and a root joined to every selector. Root plus $k + 2$ private leaves, with budget $k + 1$, gives connected domination. Selected elements can be replaced by adjacent selectors because the root already dominates all selectors. Alternatively declare the root and all elements Steiner terminals: a tree using s selectors has $n + s$ edges, so budget $n + k$ is equivalent to $s \le k$. These graphs have linear vertex and edge counts, and edge-to-vertex ratios at most four. They give the standard sparse ETH baselines for these problems. The algorithms in this paper use their explicit H_{TC} encodings.

9.5 Ancillary computations

The directory `anc/` contains finite checks of the reductions. Each check prints one summary line and writes its details to `anc/results/`. The counting checks use integer or rational arithmetic; the floating-point threshold scan is a diagnostic. Full exhaustive families and exhaustive checks on sampled supports are distinguished in Table 3. The runner executes the checks serially.

The program `hc_accounting.py` gives approximately $42.8S_G$ to $52.4S_G$ bipartite wires on its five listed sparse fixtures, with $P_0 = 4S_G + 1$. These are finite examples; the asymp-

Check	Actual finite coverage
`finite_audit.cpp`	All permutations and interval partitions through $N = 7$: 21,413,481 sign identities; 254,254 undirected marker cases through $n = 5$; 333,544 directed cases through $n = 4$
`p6_sign_check.py`	16,362 supported permutation identities; 350 sampled-support factor sums, including 218 explicit parity-wire sums
`p7_p8_filter_check.py`	120 sampled undirected graphs, 76 directed graphs; object sums and 68 metric-TSP comparisons
`e2e_hc_network.py`	Nine graph/blocking combinations, including disjoint triangles; modular readback and integer division
`check_gadgets.py`	9,000 arithmetic replays of sliced threshold/equality/modular formulas, residue hashes, and small modular-function enumeration; no tensor contraction
`check_G1_tensor.py`	3,816 consistent-input factor contractions and 411 independently assigned slice-copy cases, including full auxiliary code domains
`check_G3_tensor.py`	136,896 residue-chain factor contractions on sampled instances, all their inputs and 24 primes; invalid residue codes included
`protocol_exact_check.cpp`	Two parties, eight-bit inputs, 2,048-prime family: 3,346,110 nontrivial state/threshold classes
Explicit protocol tensor checks	Local tensors, equality subdivision, generic exact elimination, early stops and wrong-hash seeds; one extension for every original input

Table 3: Validation of the reductions. Auxiliary engine, block-count, and unused-input checks and commands appear in `anc/README.md`.

totic bounds follow from the port counts proved above. The computations do not execute the algorithm asserted by H_{TC}. Source hashes, commands, logs, and scopes accompany the programs. The arithmetic replay in `check_gadgets.py` does not test tensor-level extension counts, invalid auxiliary codes, or independently assigned copies; its modular enumeration does not implement the general integer-representation construction. The tensor checks `check_G1_tensor.py` (sliced gadgets) and `check_G3_tensor.py` (residue chains) cover the stated full-code domains. In the sliced-gadget check, independently assigned slice copies test the local gadget's extension count; the circuit's equality trees impose consistency of those copies. The ancillary numerical threshold scan is a finite scan, and the eventual inequalities are justified by the asymptotic proofs.

Use of AI

The proofs and the text were produced with AI systems (GPT-6.1 Sol, GPT-6 Astra, Claude Opus 5.5, and Claude Fable 5.1) under the author's direction. They were checked by independent reviews with AI models and by the computations in the ancillary files. No human expert has reviewed them. The author takes responsibility for the content.

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